KRITTIKA
NEWSLETTER
ASTRONOMY CLUB
IIT BOMBAY
ASTRONYX
2026
EDITION 1

THE ASTRONOMY CLUB OF IIT BOMBAY
krittika
Made of stardust.

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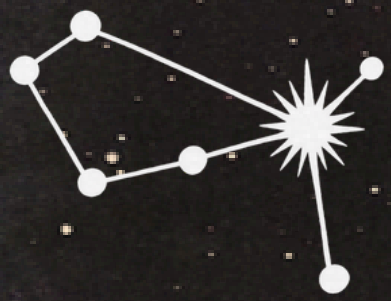


Telumendil

COSMIC NEWS

THE GREATEST COSMIC TIME LAPSE

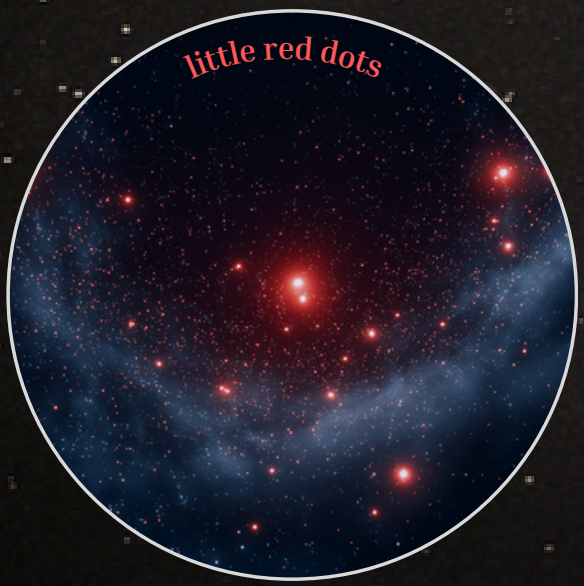
Imagine a movie that captures the entire southern sky for ten years. That's exactly what the Vera C. Rubin Observatory is creating with its colossal 3,200-megapixel camera, revealing exploding stars, racing asteroids, and cosmic events that appear and vanish in the blink of an eye. In just six weeks of testing, it uncovered more than 11,000 previously unknown asteroids, a thrilling preview of the discoveries yet to come.

Vera C. Rubin Observatory

Cosmic Time Lapse

LITTLE RED DOTS MYSTERY

For years, astronomers puzzled over how the Universe's first supermassive black holes grew so quickly. A new study suggests JWST's mysterious Little Red Dots may actually be enormous stars nearly a million times the Sun's mass, caught just before collapsing into the first supermassive black holes that seeded young galaxies.



THE VERY FIRST INTERSTELLAR SUGAR

Scientists have discovered erythrulose, the first sugar ever detected between stars, inside a giant molecular cloud near our Galaxy's centre. This finding suggests life's building blocks can form naturally in space and may have reached young planets aboard comets and asteroids, helping spark life on Earth.

INDIA'S SPACE BREAKTHROUGH

Skyroot Aerospace's Vikram-1 reached orbit on July 18, 2026, becoming India's first privately built orbital rocket. The 22-metre rocket, powered by 3D-printed engines, carried satellites into a 450-km orbit in just 15 minutes.

SKYROOT

ROMAN TAKES ON THE COSMOS

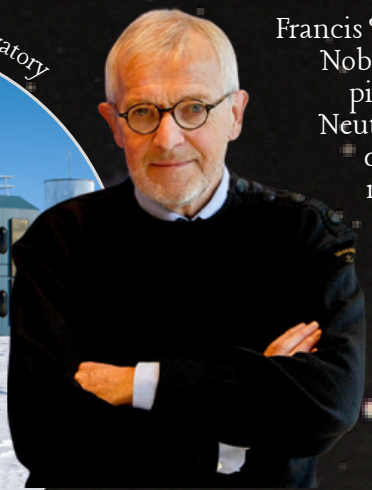
Launched in 2026, NASA's Roman Space Telescope will capture 100× more sky, map billions of galaxies, discover exoplanets, and probe dark matter and energy.

Roman Space Telescope

2026 PHYSICS NOBEL PRIZE NEUTRINOS REVEAL THE UNIVERSE

Francis Halzen won the 2026 Nobel Prize in Physics for pioneering the IceCube Neutrino Observatory and detecting high-energy neutrinos from cosmic sources, opening a new window into the distant universe.

IceCube Neutrino Observatory



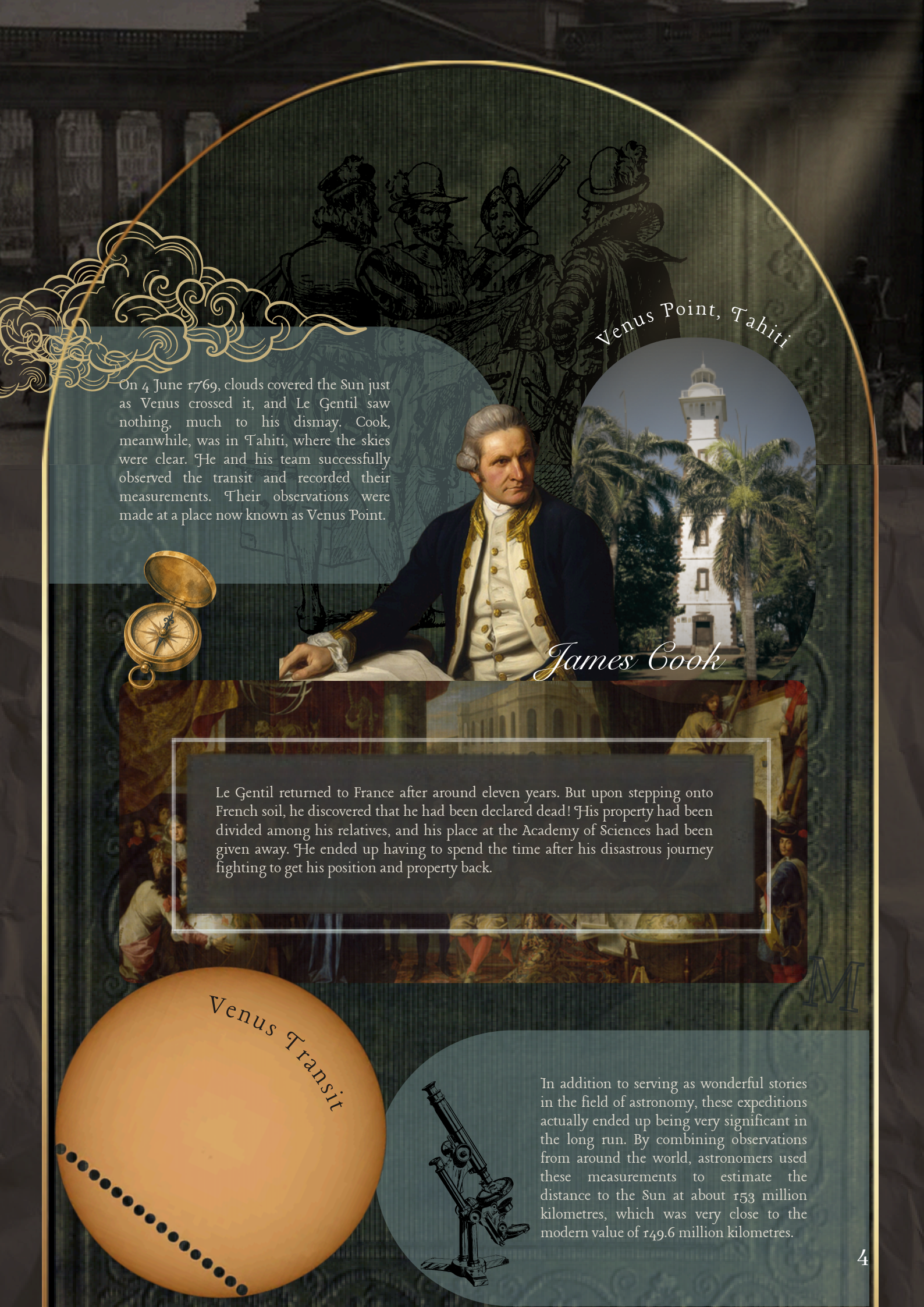
THE VOYAGE FOR VENUS

In the 18th century, astronomers were waiting for Venus to cross the face of the Sun. The transit of Venus was one of the most useful ways to calculate the distance between Earth and the Sun. We see the story of two men who set out for the 1769 transit of Venus: French astronomer Guillaume Le Gentil and British naval officer James Cook.

Le Gentil had travelled to India for the 1761 transit, only to find Pondicherry captured by the British. Stuck at sea, he could see Venus crossing the Sun, but the moving ship made his measurements useless.

Guillaume Le Gentil

Instead of going home, he made the bold choice to wait eight years for another chance at glory. He travelled around the Indian Ocean and eventually reached Manila, but Spanish authorities grew suspicious of him and treated him as a possible spy. He left for Pondicherry, built an observatory, and prepared for the next transit.



On 4 June 1769, clouds covered the Sun just as Venus crossed it, and Le Gentil saw nothing, much to his dismay. Cook, meanwhile, was in Tahiti, where the skies were clear. He and his team successfully observed the transit and recorded their measurements. Their observations were made at a place now known as Venus Point.

Venus Point, Tahiti

James Cook

Le Gentil returned to France after around eleven years. But upon stepping onto French soil, he discovered that he had been declared dead! His property had been divided among his relatives, and his place at the Academy of Sciences had been given away. He ended up having to spend the time after his disastrous journey fighting to get his position and property back.

Venus Transit

In addition to serving as wonderful stories in the field of astronomy, these expeditions actually ended up being very significant in the long run. By combining observations from around the world, astronomers used these measurements to estimate the distance to the Sun at about 153 million kilometres, which was very close to the modern value of 149.6 million kilometres.

HUNTING X-RAY BURSTS IN ASTROSAT ARCHIVES

A KRITTIKA SUMMER PROJECT

Every so often, the surface of a neutron star literally catches fire- a runaway nuclear reaction sweeps across it in seconds, briefly making it one of the brightest x-ray sources in the sky.

During KSP, a group of students used archival data from AstroSat's LAXPC instrument to find three of these bursts from a neutron star system called 4U 1705−44, and worked out what they could learn from them.

Neutron stars are some of the densest objects in the universe. They are the collapsed cores of massive supergiant stars- leftovers from the **supernova** explosion of the original star combined with gravitational collapse.

Creates an object with something like the **1.4 times the mass** of the sun packed into a sphere that on average is like 20 km across

SUPERNOVAS AND NEUTRON STARS

The result of a supernova depends largely on the core mass of the supergiant star.

2-3 times solar mass → Neutron Star

>3 times solar mass → Black Hole

RAPIDLY SPINNING PULSAR

PSR B1509−58- a rapidly spinning pulsar- can be seen in gold due to its radiation causing nearby gases to emit x-rays.

X-RAY BURSTS

Many of these neutron stars are in **binary systems**. Such neutron stars can pull massive amounts of matter from their companion star in a process known as **accretion**. There are two main mechanisms for x-ray bursts in such systems.

TYPE 1 BURSTS

Matter accumulates on the surface of the star, triggering runaway nuclear reactions.

TYPE 2 BURSTS

Violent collisions between the incoming matter and gas near the neutron star heat it to very high temperatures- emitting radiation in the x-ray region.

THE BURST CYCLE

I

ACCRETION

Matter from a companion star falls onto the neutron star.

II

BUILD-UP

Bottom layer of matter is compressed until it reaches fusion conditions.

III

RUNAWAY

A runaway reaction spreads across the star's surface- releasing enormous energy.

IV

X-RAY BURST

The star briefly produces intense x-rays- in the characteristic FRED curve.

V

COOLING

Surface cools, accretion resumes, and fuel build-up begins for the next burst.

TYPE I X-RAY BURSTS

PROJECT GOAL

Find these kinds of bursts in archival data and fit FRED models to them to detect physical parameters.

BURST ANALYSIS

We isolate individual bursts from the light curve and fit FRED models to measure key parameters. They help us understand the origin of the bursts and infer properties such as the ignition conditions, fuel composition, accretion rate, and the physical processes on the neutron star's surface. The diagram given below shows the raw actual data and on the right we show the fitted FRED model as a qualitative example for how this

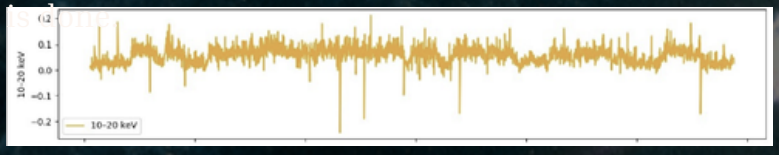


Fig. 1: Raw light curve

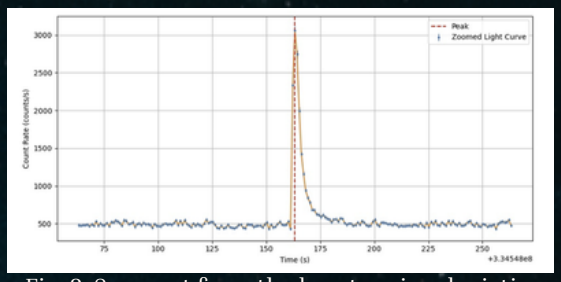


Fig. 2: Segment from the burst region depicting a 200 second window around the burst.

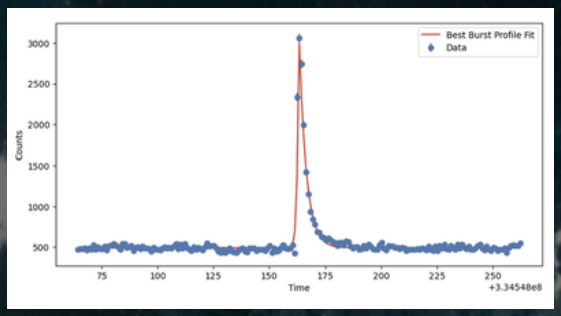


Fig. 3: FRED model fitted to the curve in question

Q&A

WITH ANIRUDH SALGUNDI

MENTOR OF THE PROJECT

Anirudh is currently a PhD student at the University of North Carolina, Chapel Hill, exploring the dynamic world of cosmic transients using massive survey telescopes like the Zwicky Transient Facility (ZTF) and the Rubin Observatory
With a strong background in studying the extreme physics of neutron stars and black hole X-ray binaries, his research focuses on uncovering the fascinating secrets of compact objects and relativistic transients.

Q **Previously you were also involved with Star Labs at IITB, what were you working on there?**

After my master's, I was looking for some research opportunities. I got into the field of X-ray binaries during my master's thesis.
I was involved with the Growth India Telescope in following up fast transients. At that time, LIGO, the gravitational wave observatory, was also on. It was sending out gravitational wave signals, and we were trying to look for electromagnetic counterparts for those gravitational waves.

Q **Do you have any advice for someone who's just starting to look into astro research on their own and is looking for research opportunities?**

The journey can start from different points in life. Some of the people I know were interested in astronomy from a very young age, some got interested once they started doing their undergrad, and some accidentally found the field.
What I would say here is, try it before saying it's not for me. I would say at least try very seriously for a couple of months to get a taste of what it is. Do start somewhere.

Q **You've been involved in research for quite some time, so would you give us an idea of what astro research looks like for someone in college who's trying to understand it in simple terms?**

Research is like doing a lot of small things at the same time. Specifically for Astro, I would say you can put them into three different boxes. One is where you're doing theory, the second where you're more observational, and the third one would be more experimental or instrumentation.
Theorists would develop new models and understanding of the phenomena. You would try to suggest a few kinds of objects that can be probed with the specifications of the telescope.

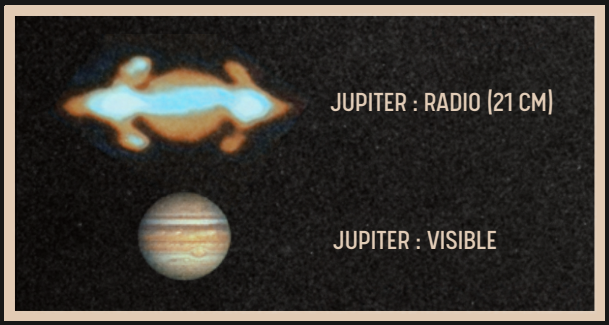
When you're doing observational work, which is mostly what I do, you are more into data analysis and developing infrastructure for finding new things in the data stream. With telescopes like Zwicky Transient Facility and Rubin LSST, the amount of data is getting exponentially higher. It's basically a huge gold mine, but someone needs to dig and find gold.
Thirdly, experimental or instrumentation. Here you would be building these telescopes, building new kinds of technology.
All in all, from what I have seen, research broadly can be put into these three buckets.
But in all three, I would say coding would at least be 30 to 40 per cent of your work.

FAST RADIO BURSTS

A MESSAGE WITH THE RETURN ADDRESS TORN OFF

CHASING MILLISECOND MYSTERIES

Monsoons in Mumbai last for a good part of the semester: something very unfortunate for us as an astronomy club. Why? Clouds. Clouds block visible light from us, or more specifically, our telescopes (it's why they look grey). This is not just a problem local to our atmosphere; optical light is easily scattered by interstellar dust and gas. This hides entire regions of the sky visible-light telescopes cannot see through.
In the meantime, radio waves have much longer wavelengths; they pass through both rain clouds and interstellar dust largely unaffected. By detecting radio waves emitted by a wide range of astronomical objects, radio astronomy reveals parts of the invisible sky.



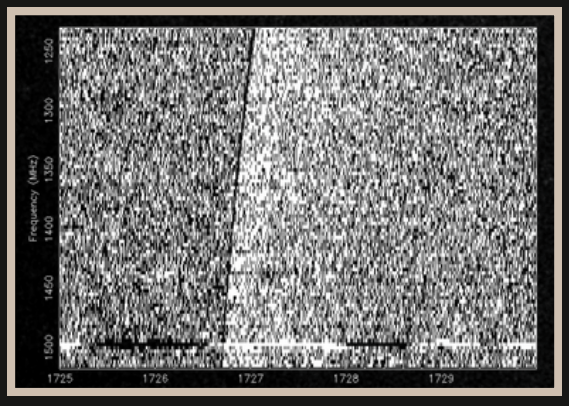
Transients are the most prized targets in radio astronomy. These sources flare briefly and usually mark a violent astronomical phenomenon happening in real time. Some well-known transients include pulsars and solar radio bursts.

One such transient lasts anywhere from a fraction of a millisecond to 3 seconds, releases as much energy as the Sun radiates in three entire days, and yet by the time it reaches Earth, the signal is 1,000 times fainter than a mobile phone call from the Moon. These are Fast Radio Bursts (FRBs), signals with great potential as cosmic probes.

TRACING THE INVISIBLE

What are these bursts? What are their sources? How can they help us map out the universe?

FRBs are extremely brief, intense flashes of radio waves from distant cosmic sources. The first, "Lorimer Burst," was accidentally discovered in 2007 by David Narkevic while studying archival data from 2001 at Australia's Parkes Observatory. 19 years and 1,000s of FRBs later, we still don't know their origin.



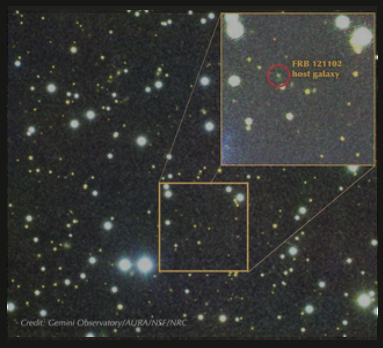
The observed DM(dispersion measure) of an FRB shows that the radio signal bears the imprint of the ionised material it traverses. They encode information about the source and its environment, the host galaxy, and our own Milky Way.

Several theories predict FRBs to be extremely potent tools. They can also hold a record of the magnetic field strength and the turbulence encountered during their passage. Their large distances make them much more intriguing, as the farther away a source is, the greater its potential for use as a cosmological probe.

Based on their observed characteristics, FRBs are the plausible outcomes of some of the most extreme processes in the Universe.

ARE THESE SOURCES ONE-TIME EVENTS SUCH AS SUPERNOVAE? OR DO THEY COME FROM ENDURING OBJECTS LIKE PULSARS THAT PERIODICALLY EMIT FLASHES?

The case of a repeating burst suggests the latter. And although we know of few repeating FRBs, it could be that all FRBs repeat and the isolated bursts represent the very brightest distribution of energies. If so, we can rule out single events and look for more persistent cosmic sources.



The leading theory is that FRBs originate from magnetars. A measurement study on FRB 20221022A was conducted in 2025 by MIT researchers. It pinned the emission close to a neutron star, favoring a magnetospheric mechanism. It was the first strong evidence directly linking an FRB to a spinning magnetar. This was backed by the fact that a magnetar, SGR 1935+2154, was caught emitting an FRB-like burst.

2007

David Narkevic accidentally discovered the Lorimer Burst at the Parkes Observatory

Matthew Bailes and *Sarah Burke-Spolaor* discovered perytons, bursts that shared many characteristics with the Lorimer bursts but were likely of terrestrial origin. Scientists begin to grow skeptical of FRBs.

2010

It was discovered that perytons were caused due to openings microwaves in the observatory before they stopped

2012

Evan Keane discovered another highly dispersed burst in archival data from the Parkes.

2015

More FRB sightings begin popping up. Telescopes capture FRBs in frequency ranges different than the Lorimer burst, providing more evidence that the burst was real.

Other theories claim that FRBs originate from mergers of neutron stars or collapsed white dwarfs in ancient galaxies. That was unexpected as magnetars are thought to form when young, massive stars explode as supernovae, so finding one in a galaxy full of old stars did not fit the usual picture. An observation made by Hubble revealed something even stranger: the FRB traced back to a cosmic neighbourhood with as many as seven galaxies on a possible path to merging, a discovery that could be significant in its own right. For a decade, catching an FRB came down to luck. Since the first one was found in 2007, radio astronomers had logged fewer than 140 bursts in more than ten years; not because these flashes are rare, but because they're almost impossible to catch in the act. An FRB lasts a few milliseconds and gives no warning.

Traditional telescopes like Parkes and Arecibo see only a sliver of sky at a time, swivelling their dishes toward one patch of the universe while the rest goes unwatched. Catching a burst meant pointing the right instrument at the right speck of sky in the right millisecond: a near-impossible bet, repeated for years.

THE CHIME REVOLUTION

Then CHIME changed the odds entirely. The Canadian Hydrogen Intensity Mapping Experiment, sitting at the Dominion Radio Astrophysical Observatory in British Columbia, doesn't swivel at all. It has no moving parts.

Four massive, stationary, cylindrical antennas simply lie in wait as the Earth rotates beneath the sky, sweeping nearly half

2016

A team led by Laura Spitler reported detecting repeated flashes from a burst. Repeating FRBs are discovered.

TODAY

More than 4,500 individual fast radio bursts have been discovered and cataloged

2018

CHIME was constructed to map hydrogen across the universe.
CHIME detected 535 new FRBs and they are published in the first CHIME/FRB Catalogue

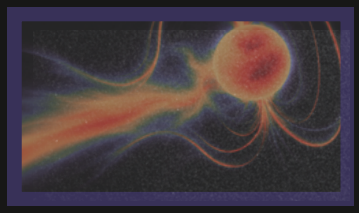
of it past the telescope's view every single day!

Instead of one receiver watching one point, a digital correlator processes incoming signals at roughly 7 terabits per second, letting CHIME watch thousands of points at once.

The results were almost absurd.
In its first year of operation alone, CHIME caught 535 new FRBs, which were more than every telescope before it, combined, in over a decade of trying.

It wasn't that FRBs were rare. They'd been going off constantly, all along, and nothing had been watching closely enough to notice. Astronomers now estimate roughly 9,000 FRBs bright enough to detect flash across the sky every day. Now the data didn't just pile up numbers - It revealed a structure nobody had seen before. CHIME found FRBs split nto two apparent groups: 18 confirmed repeaters, and a much larger set of one-off flashes that have never repeated since.

The repeaters even behave differently- with longer bursts and narrower frequency ranges, they hint towards the fact that whatever the two groups are, they might not be the same phenomenon at all but come from completely different sources.



Whether FRBs originate from distant magnetars or phenomena we have yet to discover, each fleeting flash brings us one step closer to mapping the cosmos. In a fraction of a millisecond, the deepest dark briefly gives up its secrets

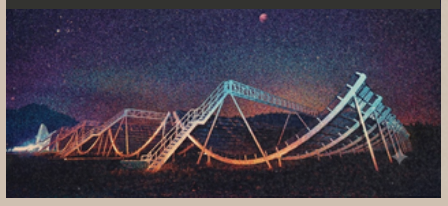
EXTRA READING

Flashes in the night: the mystery of fast radio bursts

scientificamerican.com

CHIME telescope detects more than 500 mysterious fast radio bursts in its first year of operation

news.mit.edu



Sensing the Dynamic Universe: How astronomers trace millisecond radio pulses to magnetars and distant galaxies

Hubble Finds Weird Home of Farthest Fast Radio Burst

science.nasa.gov

OBSERVATORY HIGHLIGHTS

This 279 mm of aperture gathers roughly more that **1500x more** light than the unaided human eye.

THE TELESCOPE

The Celestron C11-A is one of Krittika's COOOLEST workhorses at the observatory. It is an 11-inch Schmidt–Cassegrain with a 279 mm aperture and 2800 mm focal length, giving a focal ratio of f/10. At 12.5 kg, it is still manageable enough for a small team to set up and align without too much trouble.

Its StarBright XLT coatings help improve transmission and reduce the exposure time needed for a given target. The telescope is also HyperStar-ready, allowing the optical system to be reconfigured to f/2 for much faster imaging. 279 mm of aperture may not sound enormous — until you remember that it collects roughly 1500× more light than the unaided human eye.

ENVIRONMENTAL CONTROL

In Mumbai, humidity eats into imaging time because dew keeps forming on the corrector plate. Krittika deals with it in two ways. A dew shield extends past the front of the scope, which slows condensation and cuts stray light. A thermostat-controlled heater ring then keeps the corrector just above the dew point. Together they keep the plate dry without the heat currents that heating alone would push into the light path.

IMAGING SYSTEM

We image with a ZWO ASI2600MM Pro, a cooled mono camera built around Sony's IMX571 sensor: 6248 x 4176 px (26 MP) with 3.76 µm pixels. Two-stage cooling takes the sensor up to 35°C below ambient, and at around -20°C dark current is negligible. The camera handles up to 14 stops of dynamic range. Since it's mono, each colour channel is shot separately through a filter wheel and combined later. That takes more time than a one-shot colour camera, but the data is cleaner.

AUTOFOCUS

Focus is handled by a ZWO EAF, a stepper-driven electronic focuser with 1:128 gearing across 5760 steps. It carries imaging trains up to 5 kg without sagging, and it refocuses on its own after a filter change or meridian flip. Nobody has to touch the focuser during a session.

OPERATIONAL NOTES

At its best, Krittika's setup has managed single exposures of up to 10 minutes with round, pinpoint stars. That tells us tracking and guiding are good enough for long-exposure deep-sky work at this focal length. We also recorded transit data for a candidate exoplanet during this period. It's still unprocessed and sitting in our archive from the last observing cycle.

HOSTEL 14, IITB

ASTROPHOTOGRAPHY SHOWCASE

Deep sky captures by club members from the Whirlpool's tidal bridge to the pillars of the Eagle

Two galaxies caught mid-collision the companion NGC 5195 dragging a bridge of stars and dust from the Whirlpool's outer arm.

Whirlpool Galaxy

M51 · NGC 5194

OBJECT

TYPE	Spiral,interacting
CONST	Canes Venatici
DIST	23 Mly
PAIR	+ NGC 5195

CAPTURE

INTEG	2.6 h · 52×180s
OPTICS	SkyWatcher 130/650
CAMERA	ASI533MC Pro
SITE	GMRT · Bortle 3

A cosmic rose of ionised hydrogen, hollowed at its centre by the young cluster it gave birth to. 9.4 hours stacked across four suburban nights.

Rosette Nebula

NGC 2237 · Caldwell 49

OBJECT

TYPE	Emission · H II
CONST	Monoceros
DIST	5,200 ly
FILTER	L-eNhance

CAPTURE

INTEG	9.4 h · 188×180s
OPTICS	WO RedCat 51
CAMERA	Nikon D7200
SITE	Mumbai · Bortle 7

Pleiades

M45 · The Seven Sisters

OBJECT

TYPE	Open cluster · refl.
CONST	Taurus
DIST	444 ly
Moon	New

CAPTURE

INTEG	1.5 h · 27×200s
OPTICS	WO RedCat 51
CAMERA	ASI533MC Pro
SITE	Pahine · Bortle 4

Located 440 light-years away in Taurus, the Pleiades (Messier 45) is a culturally renowned, 100-million-year-old star cluster featuring over 1,000 stars enveloped in a blue reflection nebula, captured here under dark Bortle 4 skies.

Eagle Nebula

RACHIT AHUJA · M16 · NGC 6611

OBJECT

TYPE	Emission · H II
CONST	Serpens
FEATURE	Pillars of Creation

CAPTURE

SCOPE	Seestar S50
PROCESSING	Siril · GIMP · Starnet
SITE	Chandigarh

Stretching 4 to 5 light-years tall, the iconic Pillars of Creation are towering columns of cold gas and dust that function as stellar nurseries, simultaneously birthing newborn stars through gravity while being slowly eroded by fierce ultraviolet radiation.

A Tapestry of TRANSITS

OBJECTIVE

At first glance, a transit is deceptively simple: a planet passes in front of its star, and the star appears ever so slightly dimmer. Yet that tiny dip in brightness conceals an astonishing amount of information. For years, we have used increasingly sophisticated and sensitive machinery to detect these tiny chinks in stellar grandeur. As starlight threads its way through the planet's atmosphere, every molecule leaves behind a subtle mark, every gust of wind shifts the spectrum ever so slightly, and every rotation of the planet reveals a different slice of an alien sky.

Detecting the transit itself is a formidable challenge. Exoplanets are usually far too distant and too close to the overwhelming glare of their host stars to be imaged directly. Instead, astronomers must infer their existence from changes in a star's brightness. Such changes are often no more than a few parts in ten thousand, equivalent to noticing a fruit-fly in the distant glare of a lighthouse. Extracting the properties of an atmosphere pushes these observations to an entirely different level. During a transit, only a minuscule fraction of the starlight passes through the thin shell of gas surrounding the planet before reaching our telescopes. As it does, molecules in the atmosphere selectively absorb specific wavelengths of light, imprinting faint spectral signatures that are often buried beneath instrumental noise and the intrinsic variability of the star itself. Recovering these signals requires instruments of extraordinary precision and repeated observations.

A conventional analysis reveals only the average composition of the narrow ring of atmosphere visible during transit. This work demonstrates that even richer information is hidden within the data. As the planet moves across the stellar disc and rotates about its own axis, the exact slice of atmosphere through which the starlight passes changes continuously.

The resulting variations are exceedingly subtle, but they encode differences in temperature, chemistry, cloud cover, and atmospheric circulation across the planet. This allows us to reconstruct 3-dimensional maps of an alien world's atmosphere.

ABSTRACT

When an exoplanet passes in front of its host star in an observer's perspective (called a transit), it blocks some of the starlight. However, if the planet has an atmosphere, some of that starlight passes through the outer edges of the gas layer before reaching earth. Because different gases absorb different wavelengths of light, looking at which light gets blocked tells us exactly what the planet's atmosphere is made of. The resulting graph of the absorption v/s the wavelength absorbed is called a transmission spectrum. This study of how the light emitted from a star is absorbed and modified while passing through a transiting exoplanet's atmosphere is called transmission spectroscopy.

A light curve for a particular range of wavelengths is the brightness (flux) v/s time. The spectrum is built out of light curves for different wavelength 'bins'. When the exoplanet passes in front of the star, there is a dip in the light curve. Using the formula for flux, which depends on the area, we can calculate the ratio of the radii of the star and the planet using the magnitude of this dip. This will depend on the wavelength, since the atmosphere doesn't have the same permeability for all wavelengths: some will be absorbed, so the radius of the planet for such wavelengths will appear to include the radial extent of the atmosphere as well; some will pass through without getting absorbed, so the apparent radius of the planet will exclude that of its atmosphere.

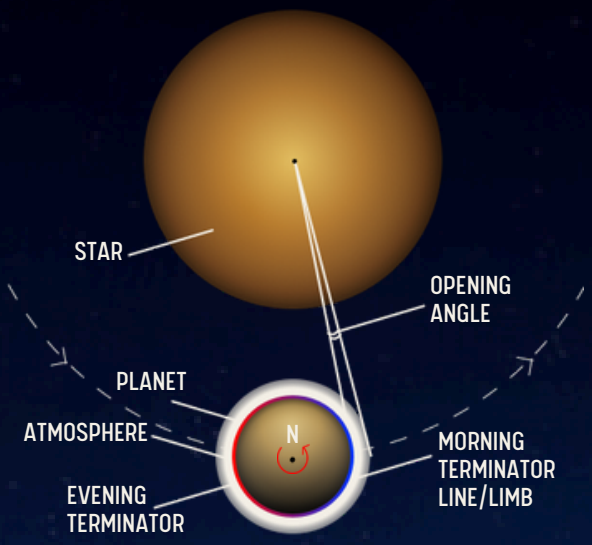
The thin ring of atmosphere around the planet that is visible to us and through which the light passes through is called the limb or the terminator line. It is essentially the boundary between day and night for the planet. The terminator ring has two distinct halves: the evening terminator, where the crust of the planet is rotating from the day toward the night-side (the sunset longitude), and the morning terminator, where the nightside gas is rotating toward the dayside (the sunrise longitude). The limb facing the direction in which the planet is revolving is the leading limb (here, the morning terminator), while that facing the opposite direction is the trailing limb (the evening terminator). The opening angle is the angular width of the atmosphere along the equator through which starlight passes. Planets rotate on their own axes as well, so the exact slice of atmosphere the light filters through keeps changing and so does the spectrum received. The universe rarely reveals its secrets directly; instead, it whispers them through light, and transmission spectroscopy is one of the most elegant ways we have learned to listen.

WASP 121-b, officially named Tylos, is an exoplanet orbiting the star WASP 121. It is an ultra-hot Jupiter: Jupiters are a type of gas-giant exoplanets that are physically similar to Jupiter, with very short orbital periods; ultra-hot Jupiters are those with a dayside temperature greater than **2200K**. In such dayside atmospheres, most molecules dissociate into their constituent atoms and circulate to the nightside where they recombine into molecules again. This planet orbits its star in **1.27** days, and is tidally locked (its rotation on its own axis and revolution around the star are synchronised such that the same side always faces towards the star or away from it). It is the first exoplanet where water was discovered.

This research uses data from the James Webb Space Telescope to dive into the finding that as this planet passed in front of and around this star, the dip in starlight (the light curve) was asymmetrical due to the planet's rotation. A normal transit light curve is symmetric in time. Since the planet's silhouette is the same throughout the transit, the star dims and brightens by the same amounts as the transit begins and ends, respectively.

A transit where the dips in the two halves of the transit are not mirror images is an asymmetric one. This asymmetry arises due to factors such as the planet's rotation, star spots, gravity darkening and instrumental glitches.

In this case, the planet's apparent size changes as the transit progresses due to its speedy rotation (relative to its revolution). Such transits are called rotational transits, where the planet rotates by a significant amount during the transit, so that the part of the atmosphere that the light filters through at the beginning doesn't overlap with that at the end. In this case, this happens because WASP 121-b is tidally locked and very close to its star. Hotter gas expands and therefore expands to the higher reaches of the atmosphere, while cooler gas stays closer to the surface. As it faces the star, the gas at the morning/ leading terminator gradually heats up and expands. So, as the hotter gas rotates into view, the trailing terminator becomes larger while the leading terminator stays smaller throughout the transit. The planet therefore appears bigger in the second half, producing the asymmetry.



A rapidly spinning host star is oblate and dimmer at its equator, a phenomenon called gravity darkening. Centrifugal force causes the star to bulge outwards at its equator, so the equatorial surface is farther away from the centre and feels less gravity. The von Zeipel relation relates the surface gravity to the surface temperature of the star:

$$T_{eff} \propto g_{eff}^{1/4}$$

where T_{eff} and g_{eff} are the effective temperature and gravity. For a blackbody, the relation between the surface temperature and the intensity of radiation emitted changes with the wavelength. As the wavelength decreases, the emitted intensity becomes exponentially sensitive to temperature. Since the planet crosses the brighter and dimmer regions during transit, this skews the light curve, but only in small amounts. The instrument **NIRISS/SOSS** aboard the **JWST** analyses slightly shorter wavelengths, while **NIRSpec** operates at longer ones, with a

small overlap in both of their ranges to ensure practical consistency. As a result, the effect decays as we move from **NIRISS** to **NIRSpec** data. The wavelengths being analysed here are such that the effects of gravity-darkening are practically negligible as compared to those of rotation.

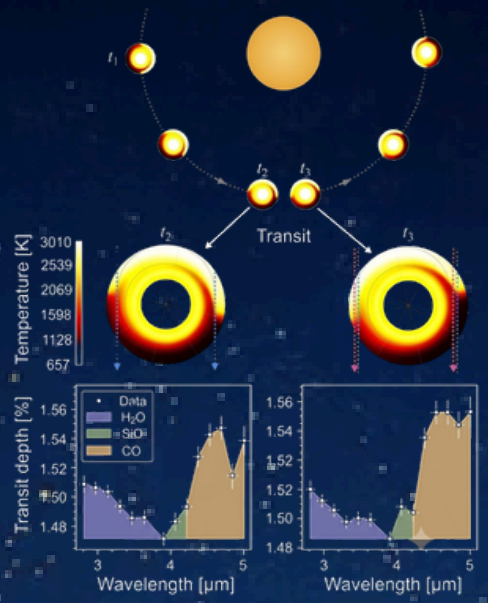
INSTRUMENTS USED

NIRSpec
(Near-Infrared Spectograph)
G395H observing mode

NIRISS
(Near Infrared Imager and Slitless Spectrograph)
Single Object Slitless Spectroscopy (SOSS) mode

OBSERVATIONS

The illustration below shows how the planet's absorbing cross-sectional area changes as the planet rotates. So, the transmission spectrum received changes with this rotational phase.



The difference in the transmission spectra at the beginning and the end of the transit reveals that the fraction of star light absorbed by the planet increases as a function of time through the entire set (or band) of wavelengths observed. This means that the planet appears to grow during the transit. Secondly, as seen in the graph, the CO absorption feature between **4.3μm** and **5.2μm** grows by **200**ppm relative to the rest of the spectrum between when the disc of the planet has just fully entered the disc of the star and when the disc of the planet is just about to leave the disc of the star. These moments when the discs of the two bodies are touching each other at exactly one point are called contact points and there are of them in most transits. Here the increase in CO absorption was measured between contact points and . In the same interval, the H_2O feature remained approximately constant.

As the planet rotated from west to east during the transit, CO absorption in the spectrum increased while H_2O absorption decreases. A longitudinal temperature gradient denotes the change in temperature as one moves from east to west along the planet's surface. The study observed that the change in temperature (local temperature gradient) was higher at the evening terminator than the morning one. The temperatures in the eastern half are higher than in the western half of the dayside.

DATA ANALYSIS

Regarding longitudinal gradients of molecular abundances, the Doppler shifts of CO and H_2O lines in **WASP 121-b** were previously measured as functions of orbital phase during three transits observed using the Immersion Grating Infrared Spectrometer **(IGRINS)** on the Gemini-South Telescope. The data revealed CO lines to become increasingly blueshifted during the transits indicating that CO absorption predominantly originates in the trailing limb that rotates toward the observer. The blueshift of H_2O lines, in contrast, decreased as the transits progressed, revealing H_2O absorption to be dominated by the leading limb that rotates away from the observer. These results are in line with the **JWST** observations presented here that also suggest CO, but not H_2O, absorption to trace the temperature increase in the trailing limb as the planet rotates during transit due to the thermal dissociation of H_2O.

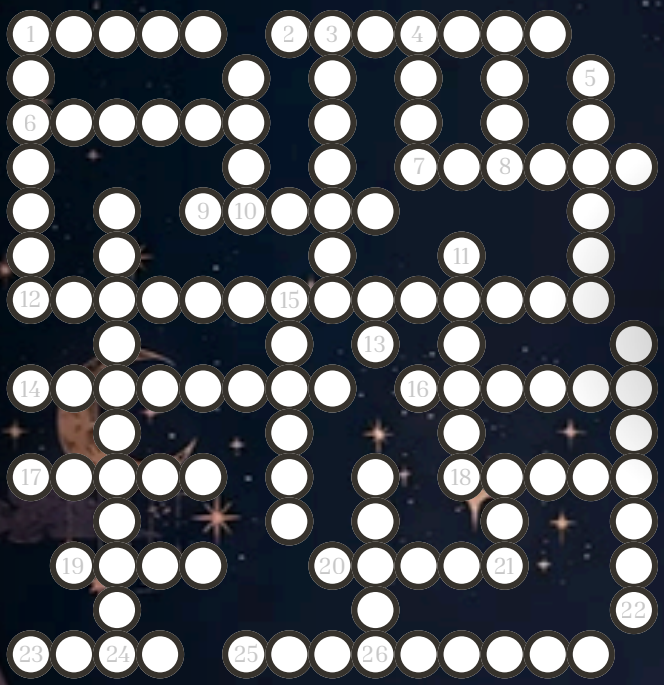
CONCLUSION

Both the **NIRISS** and the **NIRSpec** data point toward increasing CO absorption in the planet as a function of orbital phase with quantitatively consistent amplitudes. This demonstrates **JWST's** remarkable capability to measure phase-dependent absorption in transiting exoplanets consistently between different observations and instruments.

HOROCRUX

A CRYPTIC CROSSIE

This puzzle needs not merely an eye for stars but a complementary flair for wordplay. Warp your brains and contort the clues to arrive at words with astronomical significance. If you succeed, share a picture of your completed grid on instagram and tag @krittika.iitb.
The first few solvers get a chance to be rewarded at Astromania!



ACROSS:

1. Frequent heavenly visitor, come tails off (5)
2. A tidy list from an untidy astronomer (7)
6. Ringed planet, runs at chaos (6)
7. Elements clouding atop a new boulevard (6)
9. A philosopher disemvowelled (5)
12. Poke your nose around, the study (14)
14. Sounds like a group of 3 new card game packets (9)
16. Twin twinklers, can mess up your horrid pictures (6)
17. Doesn't Al Gore sound like the less favoured twin? (5)
18. Leads the nearest horseman pack (5)
19. Appearances spoilt by the lord of war (4)
20. Ram yourself into the warlord (5)
23. Bear-ish sentiments large and small adorn the northern sky (4)
25. Call the first part, halving the range (9)

DOWN:

1. The star squad initially looked like it'd bag a W, but it broke off and caved in (7)
3. A cue waiter's continuous speech divides the planet (7)
4. A bird flies southwest, initially national air (4)
5. A chocolate, a whirl (6)
11. Soviet probe, venerated at the beginning (6)
15. Yelling in the middle, thanks without a kiss, signal the start of a messy relationship (6)

UP:

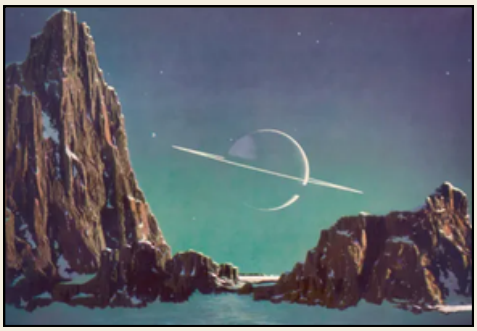
8. A grand 9-squared design to augur well without the stars (4)
10. Orbiter commences madness (4)
13. The last twain of old Macdonald's ditty orbit jupiter (2)
21. Martian solid in liquid (3)
22. Orients us towards a misdirected thorn (7)
24. Celebs form a bound heavenly companionship (4, 7)
26. Home is where the heart lies messy (5)

DOTTING THE EXPANSE

Something's hiding in plain sight,
like stars in broad daylight.
You must find the hidden entities in this star-studded magazine.
Beware, though, that all that you perceive as a star in the sky might not truly be one..

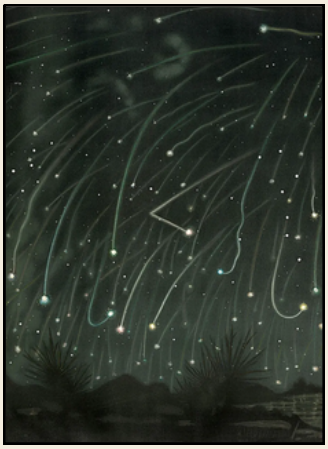
Count the hidden objects and post the number onto your Instagram story as <name of the object> <number of copies found>, and make sure to tag @krittika.iitb.
Few correct answers stand a chance to win freebies !!

MUSES &



Saturn as seen from Titan
Chelsey Bonestell, 1944

Titan is Saturn's largest moon and the second-largest moon in the entire solar system. Although we've learned that the view from Titan would look much different than seen here, Bonestell's painting was very accurate for its time.



November Meteors
Étienne Trouvelot, 1868

In the freezing early morning hours of November 14, 1868, Trouvelot locked himself by an upstairs window at his home. From midnight to 5:00 AM, he sat shivering in the dark, completely spellbound by a massive, steady stream of meteors cutting through the clear black sky. He accurately depicts a 'radiant point' in the sky where a heavy bombardment of light trails falls toward the ground.

The Appearance of Halley's Comet
Bayeux Tapestry
11th century, Bayeux Museum

Back in the 11th century, comets were treated as ominous. Medieval artisans embroidered this visual story using needle and thread. They creatively manipulated the timeline, stitching the comet right after King Harold's January coronation rather than its actual appearance four months later, framing it perfectly as a dramatic, doom-bringing plot device. In this 11th-century embroidered comic strip, Halley's Comet serves as a divine eviction notice.

Drawing of Messier 51
William Parsons, the Third Earl of Rosse, 1845, compared to a more modern photograph.

Before astrophotography, people drew galaxies by hand. The object was the first 'Spiral Nebula', i.e., a deep-sky object where a spiral structure was detected. This demonstrates how accurately Lord Rosse recorded the finest details in the spiral arms of this galaxy.

MUSINGS

Forty-two years ago (to me if to no one else
The number is of some interest) it was a
brilliant starry night
And the westward train was empty and had
no corridors
So darting from side to side I could catch the
unwonted sight
Of those almost intolerably bright
Holes, punched in the sky, which
excited me partly because
Of their Latin names and
partly because I had
read in the textbooks
How very far off
they were, it seemed
their light
Had left them (some
at least) long years
before I was.
And this remembering now
I mark that what
Light was leaving some of them at least then,
Forty-two years ago, will never arrive
In time for me to catch it, which light when
It does get here may find that there is not
Anyone left alive
To run from side to side in a late night train
Admiring it and adding noughts in vain.

This poem strands you. It takes you down the road of the poet's nostalgia while he contemplates the sheer distance between him and the stars. How, when the light emitted by the stars 42 years ago finally reaches Earth, he wouldn't be there to see it. And maybe no one else will.

~Louis MacNeice
1963

Give it a listen and you might just find yourself grooving to some upbeat Apollo 11 mission control audios.

Go!
Public Service
Broadcasting
2015

The song follows a fictional astronaut, Major Tom as he prepares for a lonely death in space at the face of a malfunctioned spacecraft.

Space Oddity,
David Bowie
1969

A very rhythmic song that follows an aspiring astronaut observing- you guessed it- Kepler-22b with a cheeky plot twist.

Kepler-22b,
King Gizzard &
the Lizard
Wizard, 2022

CINEMA BEYOND SPACE

MOON

The film explores the lonely journey of astronaut Sam Bell. A man who experiences a personal crisis as he begins to experience hallucinations and physical ailments nearing the end of a three-year solitary stint on the far side of the moon.

SOLARIS

An eerie adventure that explores a psychologist's journey to a station orbiting a distant planet as he discovers the entire crew has gone insane.

CONTACT

A signal is detected on Earth from a foreign planet instructing the humans to build a one man spaceship using science which no one on Earth can understand. The film by Robert Zemeckis brilliantly explores human reactions on something they can't understand.

**We have restrained ourselves from choosing the most basic movies like Interstellar, Gravity, The Martian and Arrival. And to whoever was mailing us constantly, NO! We are not including "Koi Mil Gaya" in this.*

October Sky

In a different turn from our remaining movies, the film is the sweet and inspirational story of Homer Hickam Jr. and shows how the son of a coal miner dreams of studying rocketry and achieves the dream against the society as well as his father's wishes.

2001

a space odyssey

Released one year before the actual moon landing, the film is the most accurate depiction of space in any movie and the visual effects hold up more than half a century later.

GATTACA

A man tries to defy a system obsessed with genetic perfection as he assumes the identity of a genetic elite being "genetically-inferior" himself to pursue his goal of travelling into space.

ALUMNI REFLECTIONS

Gaurav Waratkar, a Postdoctoral Fellow at Caltech's Cahill Center for Astronomy & Astrophysics, has been a part of Krittika for over a decade. Gaurav has contributed to a plethora of fascinating projects, including but not limited to the Growth India Telescope, Daksha (Proposed satellite by STAR lab, IITB and other institutes), Seismometers (at LIGO) and many many more, combining his love for mechanical engineering and astronomy. He currently works on astronomical instrumentation at Caltech.

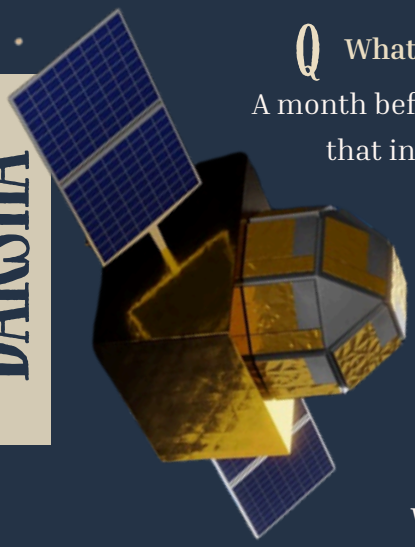
Q You have been involved with Krittika for over 10 years; how has your experience with the club been?

I try to stay away, but I can't; it's been really good.
I started in my first year. What stood out to me most about the club was the cool seniors and the community. Back then, I attended every event and absorbed as much as I could.
As a convener, I saw the club through a different lens. As a manager, I saw it from another. Now, post-manager, I see the club from a new vantage point altogether.
When I joined, Krittika was in the middle of a transition. Certain initiatives we took really shot up the kind of reach, the kind of impact, and the importance of the club. Over the past decade, it has grown tremendously.
Over the years, I deliberately stayed in touch because I got something from my seniors, and I am giving that back to the folks who need a flavour of what Astronomy is.
There have been ups and downs over these 10 years. Growth happens outside your comfort zone, and Krittika stepped outside its own, sometimes through failures. But we learned from those experiences, and it's been great.

Q What are the courses and resources that helped you pursue astronomy?

Astronomy is a very interdisciplinary topic. You find Physics, different aspects of physics; you find a lot of engineering; you find a lot of software. There's the standard Introduction to Astronomy course and courses like Introduction to gravitational waves, simulations, stellar evolution, etc. All these courses are exceptionally good and really important if you want a flavour of what research actually means. Some courses give you a fundamental understanding of processes that happen in Astronomy that eventually form your observables, like radiation processes, atomic and molecular physics, nuclear physics, etc. All extremely important. In software, standard minor courses in CS and CMinds give you skill sets you can apply to a data-hungry field like Astronomy.

DAKSHA



Q What sparked your interest in astronomy, and when did it start?

A month before my JEE Advanced, I started getting distracted by things that interested me. One of my teachers had given me a book about observing the night sky, and exactly a month before JEE, I happened to open it. I went out to my terrace, looked up at the sky, and spotted constellations! So I was like, okay, this is cool. I need to pay attention to this. I then started reading books and got hooked. It wasn't a serious career prospect until I entered IITB. In the first few weeks, we had that fantastic STAB orientation, or ITC orientation. We called it STAB back then. Krittika literally blew me away. That was the start of my Krittika adventures, which have continued since 2015. In my first year, I attended every Krittika event. Even the conveners did not attend as many events as I did. And at the end of my first year, I officially joined Krittika as a convener. When that happened, I got in touch with seniors who had careers in Astronomy.

Q What advice would you give to students interested in your field?

Start by stepping outside. Astronomy is one of the most intuitive and accessible fields. One of the most important questions that humans try to answer is one of those fundamental questions, right? Who are we, and so on. Those really work out when you go and see yourself as a speck of dust in the entire universe. And that only happens when you go outside your bubble. Step out, see the stars, observe them. That's all. Then see where you want to go from there. Talk to people, try to understand what they're doing.

Speak to your seniors to be sure that it is going to be useful for you, and it's going to be worth your time; otherwise, you may find that it is something that you would like to know, but it is possible that it may not account for something actually useful in the sense that you want to do in the future. I chose Astronomy as something I would give myself a year to figure out, and then, if I felt like it was fun from a distance and not really something I liked, I would pivot back. But I really liked it, and that started my Astronomy career. .

Krittika Trips

January

Take astronomy beyond the campus

Travel to new places, chase darker skies, and experience the cosmos through stargazing, exploration, and shared adventures.

National Space Day

SOM Foyer 22nd August

Celebrate India's journey beyond the earth

A day of exhibitions, talks, screenings, and hands-on experience that brings the wonder of space exploration closer to everyone.

Movie Screenings

See the universe through a different lens

A cinematic escape into space, followed by conversations that make you look at what you just watched a little differently.

KSP/KCAP/LS

Sem Breaks

Go from learning astronomy to actually exploring it

Research projects, computational challenges, and structured learning experiences that take you from curious beginner to hands-on explorer of the cosmos.

KRIT CHRON

TIKA

!

tika

ICLES

Astromania

LA001 14th Oct

THINK YOU KNOW THE COSMOS? ASTROMANIA PUTS IT TO THE TEST

A high-energy astronomy quiz packed with unexpected questions, rapid-fire rounds, and twists that separate the stargazers from the space nerds.

Observatory

Hostel 14

WHERE THE UNIVERSE COMES A LITTLE CLOSER

Hands-on access to astronomical instruments for exploring the night sky, observing celestial objects, and capturing them through astrophotography.

Friday Night Stargazing

Gymkhana Grounds

LOOK UP! THERE'S A LOT MORE THAN YOU THINK

Guided night-sky sessions where you can use telescopes to explore the cosmos firsthand and learn to navigate the stars for yourself.

Talks & Workshops

Physics Dept. Seminar Hall

WHERE THE UNIVERSE BECOMES A CONVERSATION

Engaging talks and hands-on workshops that turn complex ideas into something you can explore, question, and experience.

KRITIKA

THE SUPER SQUAD

JAHNAVI

VIHANG

ABHINAV

ADITYA

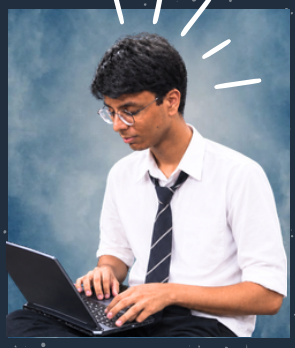
AKSHIT

ANISHA

ARHAM

AYUSHI

ISHAAN



JATIN

MIHIKA

PARAS

PRACHI

RISHABH

RITVIK

SIDDHI

SRUSHTI

UDDIPTA

"per aspera,ad astra."

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Until the stars align again.
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